4.1

In [30]:
```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.decomposition import LatentDirichletAllocation

count_vectorizer = CountVectorizer(max_features = 2000, min_df = 1, max_df = 0.9)
X_counts = count_vectorizer.fit_transform(articles_data["Clean_Content"])
print("Document-term matrix shape:", X_counts.shape)

n_topics = 4
lda = LatentDirichletAllocation(n_components = n_topics, random_state = 42, max_iter = 100)
doc_topic = lda.fit_transform(X_counts)

feature_names = count_vectorizer.get_feature_names_out()
n_top_words = 8
topic_word_lists, topic_weight_lists = [], []
for topic in lda.components_:
    top_idx = topic.argsort()[::-1][:n_top_words]
    topic_word_lists.append([feature_names[i] for i in top_idx])
    topic_weight_lists.append([topic[i] for i in top_idx])
    print("Topic", len(topic_word_lists), ":", ",".join(topic_word_lists[-1]))
```

```
Document-term matrix shape: (5, 2000)
Topic 1 : alpha,development,instrument,cronbach,science,reliability,scale,design
Topic 2 : method,triangulation,mixed,design,multiple,denzin,quantitative,source
Topic 3 : recognition,move,becoming,audit,attention,network,axiological,ontological
Topic 4 : reflexivity,position,self,effect,personal,participant,meaning,reaction
```

In [31]:
```
doc_topic_data = pd.DataFrame(doc_topic,
                              columns = [f"Topic {i + 1}" for i in range(n_topics)],
                              index = articles_data["filename"])

print("Document-topic distribution:\n")
print(doc_topic_data.round(3))
print("\nDominant topic per article:")
print(doc_topic_data.idxmax(axis = 1))
```

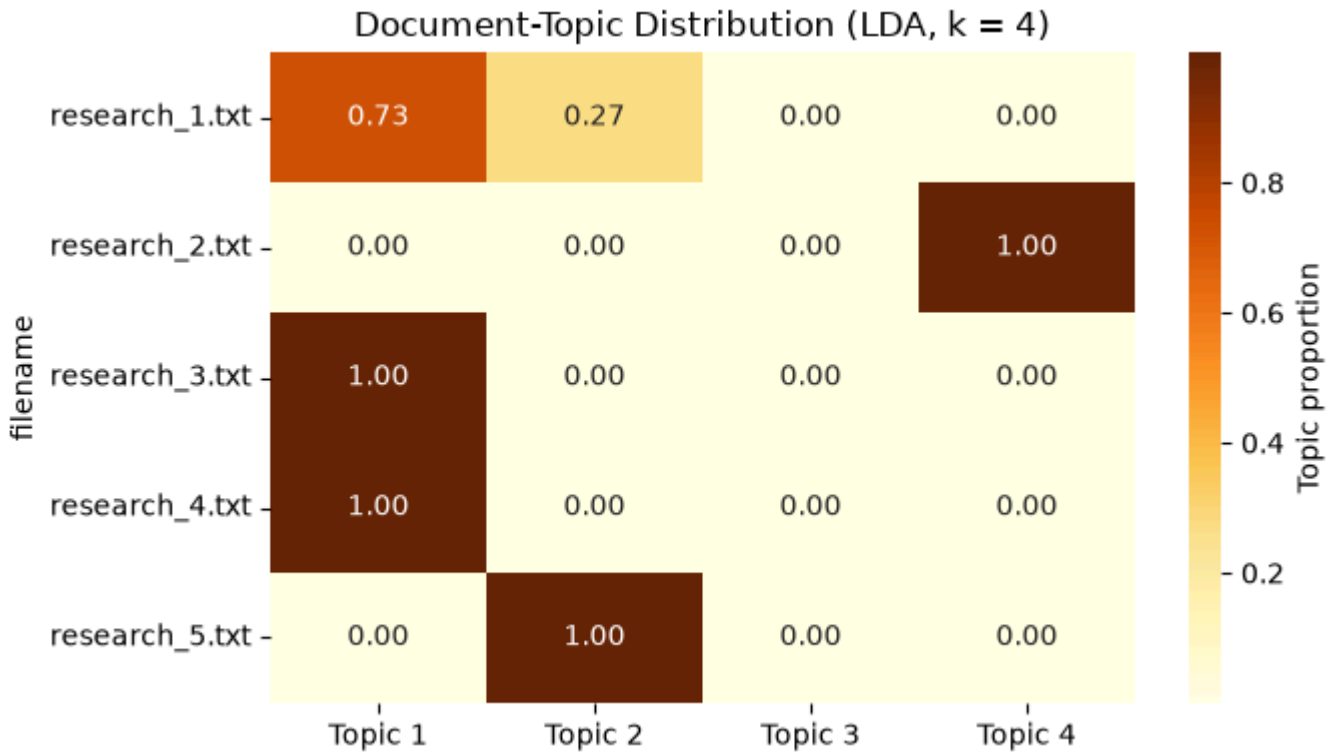
```
Document-topic distribution:

               Topic 1  Topic 2  Topic 3  Topic 4
filename
research_1.txt    0.728    0.271      0.0    0.000
research_2.txt    0.000    0.000      0.0    0.999
research_3.txt    1.000    0.000      0.0    0.000
research_4.txt    1.000    0.000      0.0    0.000
research_5.txt    0.000    1.000      0.0    0.000

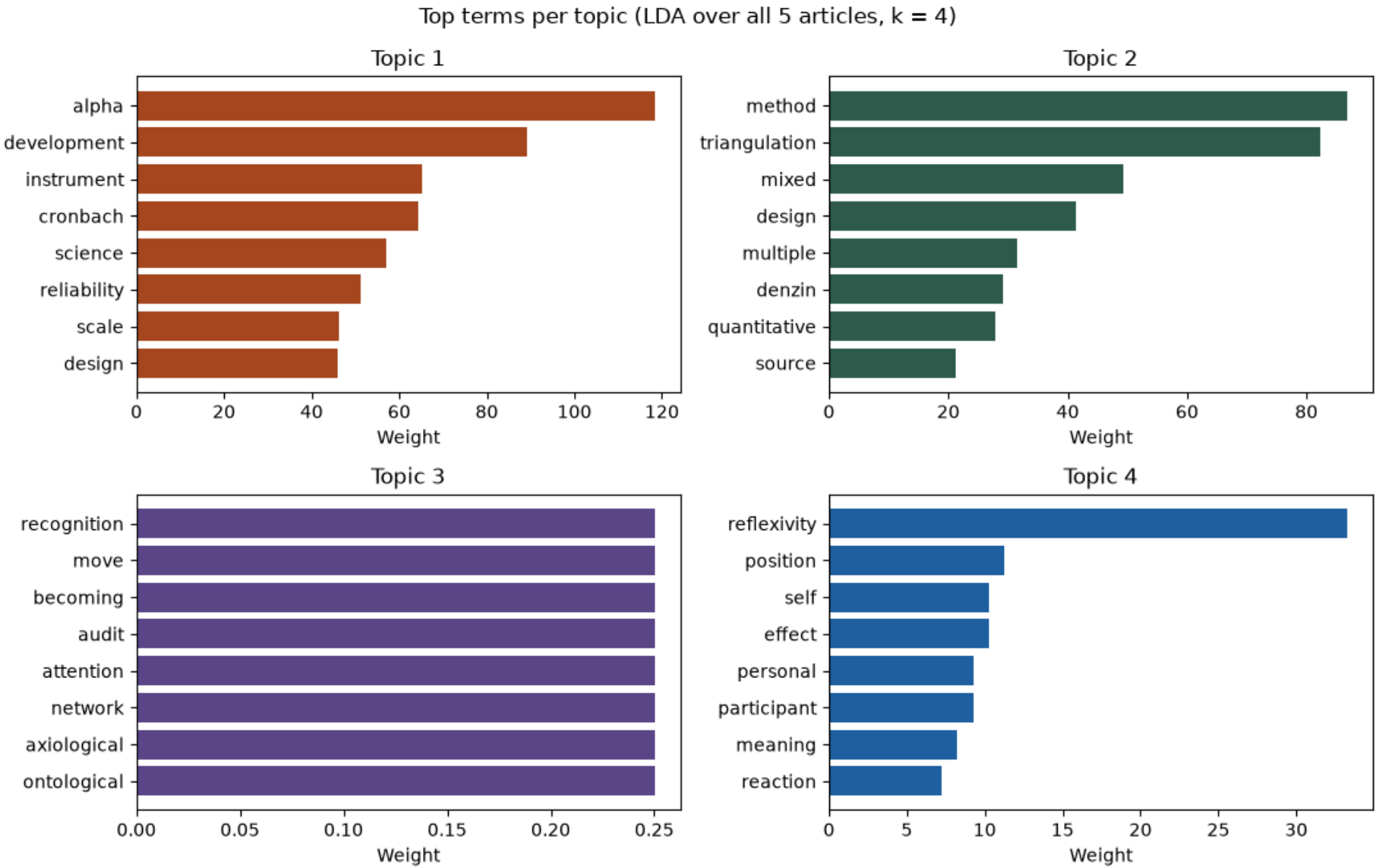
Dominant topic per article:
filename
research_1.txt    Topic 1
research_2.txt    Topic 4
research_3.txt    Topic 1
research_4.txt    Topic 1
research_5.txt    Topic 2
dtype: str
```

In [32]:
```
import seaborn as sns

fig, ax = plt.subplots(figsize = (7, 4))
sns.heatmap(doc_topic_data, annot = True, fmt = ".2f", cmap = "YlOrBr", cbar_kws = {"label": "Topic proportion"}, ax =ax)
ax.set_title("Document-Topic Distribution (LDA, k = 4)")
plt.tight_layout()
plt.show()
```



In [33]:
```
fig, axes = plt.subplots(2, 2, figsize = (11, 7))
colors = ["#a8481f", "#2f5d4f", "#5b4a8a", "#2060a0"]
for i, ax in enumerate(axes.flat):
    words = topic_word_lists[i][::-1]
    weights = topic_weight_lists[i][::-1]
    ax.barh(words, weights, color = colors[i])
    ax.set_title(f"Topic {i + 1}")
    ax.set_xlabel("Weight")
plt.suptitle("Top terms per topic (LDA over all 5 articles, k = 4)")
plt.tight_layout()
plt.show()
```



4.2

In [36]:
```
import pyLDAvis
import pyLDAvis.lda_model
```

```
panel = pyLDAvis.lda_model.prepare(lda, X_counts, count_vectorizer, mds = "tsne")
pyLDAvis.save_html(panel, "lda_vis.html")
print("saved interactive map to lda_vis.html")

panel
```

```
saved interactive map to lda_vis.html
```

Out[36]:
```
PreparedData(topic_coordinates=               x          y  topics  cluster       Freq
topic
0      110.803604  -39.273762       1        1  65.178568
1       -1.484537   61.546577       2        1  23.829302
3     -102.400642  -50.472221       3        1  10.975843
2        9.718333 -150.793808       4        1   0.016286, topic_info=              Term        Freq       Total Category  l
ogprob  loglift
1924  triangulation   65.000000   65.000000  Default  30.0000  30.0000
1169         method   76.000000   76.000000  Default  29.0000  29.0000
1193          mixed   39.000000   39.000000  Default  28.0000  28.0000
60            alpha  107.000000  107.000000  Default  27.0000  27.0000
1595    reflexivity   21.000000   21.000000  Default  26.0000  26.0000
...             ...         ...         ...      ...      ...      ...
1545      rationale    0.000626    3.025174   Topic4  -7.6009   0.2401
1548        reached    0.000626    3.025174   Topic4  -7.6009   0.2401
1566     recognized    0.000626    3.025174   Topic4  -7.6009   0.2401
1634         remain    0.000626    3.025174   Topic4  -7.6009   0.2401
1919          treat    0.000626    3.025174   Topic4  -7.6009   0.2401

[246 rows x 6 columns], token_table=      Topic      Freq           Term
term
4         3  0.826775        absence
7         3  0.826775         abused
10        3  0.826775         accent
13        1  0.997783     acceptable
14        3  0.826775     acceptance
...     ...       ...            ...
1924      3  0.015236  triangulation
1954      1  0.296844       validity
1954      2  0.692636       validity
1967      1  0.967799         volume
1983      3  1.004316          woman

[320 rows x 3 columns], R=30, lambda_step=0.01, plot_opts={'xlab': 'PC1', 'ylab': 'PC2'}, topic_order=[1, 2, 4, 3])
```

4.3

https://github.com/T-RoyMC/ITNPA4-B34

In [ ]: